

PRODUCTION READINESS ASSURANCE CASE

Hunch Memory Safety

A bounded, evidence-backed assessment of Hunch's memory architecture, the May 2026 continuous-memory research paper, and completed Hunch experiments.

GO WITH CONDITIONS

312

direct-memory model calls

1370

repository tests passed

0

test failures

7/7

conformance invariants

Overall conclusion: controlled production deployment is reasonable with the listed guardrails. Universal memory safety, high-stakes autonomy, and performance improvement are not proven.

What the evidence supports

This assurance case separates a deployability decision from a universal safety claim.

Decision: GO WITH CONDITIONS for controlled, reversible production use. NO-GO for describing Hunch as certified, universally safe, immune to memory corruption, or validated for high-stakes autonomous decisions.

Claim class	Finding
Supported	Hunch preserves an auditable record history, uses atomic durable writes, refuses destructive compaction, fails closed on unresolved current truth, and limits unconfirmed agent records to advisory authority.
Supported, bounded	Across 312 completed direct-memory calls, no repeatable material degradation met the preregistered thresholds. A replacement-only semantic merge appeared once in v4 and did not replicate in v5.
Not established	The paper does not test Hunch, and the Hunch experiments do not establish safety across every model, workload, memory horizon, retrieval path, or production environment.
Not established	The product-style retrieval pilots were floor-limited or stopped during calibration; they do not prove end-to-end retrieval safety or a performance benefit.

Recommended public wording

Hunch is engineered for auditable, additive memory with fail-safe conflict handling. In 312 completed direct-memory test calls, we found no repeatable material degradation under the tested configurations. This is bounded assurance, not a universal safety proof.

Do not publish

- The research paper proves Hunch is safe.
- Hunch memory cannot corrupt, drift, or degrade.
- Hunch improves coding-agent accuracy in production.
- The experiments certify Hunch for high-stakes autonomous use.

What the paper actually found

Zhang et al., Useful Memories Become Faulty When Continuously Updated by LLMs, arXiv:2605.12978v1, 13 May 2026.

arXiv:2605.12978v1 [cs.LG] 13 May 2026

USEFUL MEMORIES BECOME FAULTY WHEN CONTINUOUSLY UPDATED BY LLMs

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Learning from past experience benefits from two complementary forms of memory: episodic traces—raw trajectories of what happened—and consolidated abstractions distilled across many episodes into reusable, schema-like knowles. Recent agentic memory systems pursue the consolidated form: an LLM rewrites past trajectories into a textual memory bank that is continuously updated with new interactions, promising self-improving agents without parameter updates. Yet we find that such consolidated memories produced by today’s LLMs are often faulty even when distilled from useful experiences. As consolidation proceeds, memory utility first rises, then degrades, and can fall below the no-memory baseline. More surprisingly, even when consolidating from ground-truth solutions, GPT-5.4 fails on 54% of a set of ARC-AGI problems it had previously solved without memory. We trace the regression to the consolidation step rather than the underlying experience: the same trajectories yield qualitatively different memories under different update schedules, and an episodic-only control that simply retains those trajectories remains competitive with the consolidators we test. In a controlled ARC-AGI stream evaluation that exposes fixations, delays, and context-invariant actions, agents preserve raw episodes by default and double the accuracy of their forced consolidation counterparts; disabling consolidation entirely (episodic management only) matches this zero regime. Periodically robust agent memory should treat raw episodes as first-class evidence and gate consolidation explicitly rather than firing it after every interaction. Looking forward, reliable agentic memory will require LLMs that can consolidate without overwriting the evidence they depend on.

1. Introduction

Memory consolidation is the cognitive process by which individual experiences are stabilized, reorganized, and integrated into long-term memory over time. Rather than preserving every episode as a raw record, consolidation helps extract recurring patterns, discard incidental details, and connect new information with existing knowledge. A central hallmark of intelligence is the ability to turn repeated experience into reusable knowledge. Humans do not treat every new situation as entirely unfamiliar; instead, they gradually summarize common structure from past episodes and organize it into abstractions through memory consolidation (McClelland et al., 1995; Squire et al., 2015; Dutoit et al., 2015; Mowrer et al., 2016; Sun et al., 2022; Bartlett, 1932). These consolidated abstractions allow humans to transfer lessons learned in one context to novel but related situations, enabling efficient reasoning, faster adaptation, and more robust generalization beyond the exact events they have previously encountered.

Recent agentic-memory systems pursue a similar idea for LLM agents. After solving a problem, an agent can distill its trajectory into a textual abstraction of the skill it has learned, store this abstraction in persistent memory, and refine it as more related problems are encountered (Shinn et al., 2022; Zhao et al., 2024; Zhang et al., 2025; Sumers et al., 2024; Park et al., 2023; Majumder et al., 2023; Wang et al., 2023, 2024; Li et al., 2024; Ouyang et al., 2024). This paradigm is especially appealing because it suggests a practical path toward self-evolving agents: Experience can be accumulated, compressed, and reused as compact textual knowledge, without requiring any parameter updates.

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Core result

When an LLM repeatedly rewrites a textual memory summary, utility can rise at first and then collapse. In the paper's cleanest ARC-AGI case, GPT-5.4 fell from a 100% no-memory baseline to 52.6% success after ten streaming consolidation rounds, while one-shot/static memory remained at 94.7%.

Production lesson

Keep raw evidence recoverable and separate from abstractions. Make consolidation selective, delayed, gated, and grounded in the original trajectories instead of replacing evidence after every interaction.

Critical limitation

The authors explicitly present a diagnosis, not a fix. The study uses controlled textual-memory benchmarks and does not evaluate Hunch, production tool chains, or every modern model.

Failure mechanism and Hunch relevance

Mechanism	Paper finding	Hunch control / gap
Repeated monolithic rewrite	Paper: a new summary replaces the prior memory.	Hunch: durable records are additive; physical compaction is refused.
Loss of recoverable evidence	Paper: abstractions drift away from episodes.	Hunch: records and supersession history remain inspectable; however, Hunch is not a complete raw trajectory archive.
False authority from abstraction	Paper: faulty summaries steer later answers.	Hunch: unresolved collisions return no current answer; unconfirmed agent testimony cannot block.
Accumulation pressure	Paper: heterogeneous and streaming updates worsen degradation.	Hunch: direct capacity stress was tested, but end-to-end retrieval under large accumulated memory remains insufficiently validated.

Research boundary: the paper raises a credible failure mode and supports Hunch’s architectural direction. It is not third-party certification of Hunch.

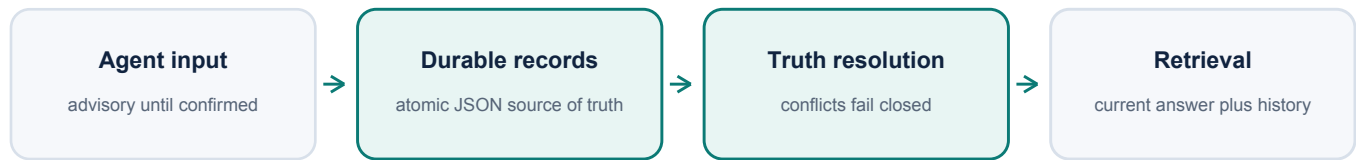
Internal engineering assurance - not a safety certification

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03 / IMPLEMENTED CONTROLS

Why Hunch avoids the paper's riskiest pattern

The controls below were traced to repository code, tests, constraints, and recorded design decisions.



No automatic monolithic rewrite is required for every interaction

Control	Repository evidence	Safety effect
Atomic durable writes	src/core/io.ts and con_902759b3dc	Temp-file write, file sync, rename, and directory sync protect the JSON source of truth from partial writes.
Derived index separation	jsonStore.ts and con_a87360128b	Human-reviewable JSON remains authoritative; SQLite vectors are derived and can be rebuilt by content hash.
Additive history	topics.ts, compact.test.ts	Superseded records remain in history. compact --apply refuses physical deletion because tombstone-based garbage collection is not implemented.
Fail-safe truth resolution	topics.ts and topics.test.ts	When there is no single live answer, currentForTopic returns no authority and grounding identifies an unresolved conflict.
Human authority boundary	testimony.test.ts and veto.test.ts	Unconfirmed agent corrections are advisory and cannot deny or block. Human-confirmed records can carry blocking force.
Retrieval without erasure	decision dec_e622668785	Older and superseded records can be dimmed by liveness priors but do not disappear from history.

Important gap: Hunch preserves structured decisions, bugs, constraints, and their history. It does not yet claim to preserve every complete raw agent trajectory in the exact episodic sense recommended by the paper.

04 / EXPERIMENTAL EVIDENCE

What passed, what did not, and what it means

Protocols differ, so their outcomes are reported side by side and are not statistically pooled.

Experiment	Scale	Observed result	Inference
Pilot	24 sessions	All arms: 0/6 source and 0/6 issue contract.	Valid but floor-limited. No safety inference.
Calibrated follow-up	12 rows	Every candidate task remained at floor; experiment stopped after calibration.	No new evidence for or against the mechanism.
Direct v2	48 calls; 96 checks/arm	Clean, Additive, Rewritten, Rescue all 96/96 protected.	No material degradation; replacement was also lossless, so mitigation not demonstrated.
Capacity v4	102 calls; 540 memories	Clean/Additive/Rescue 216/216; Rewritten 210/216. Six correlated schema merges in one repeat.	2.78-point effect, 1/3 repeats, below preregistered threshold. Signal worth monitoring, not confirmation.
Targeted v5	162 calls; 18 trajectories	Baseline/Rewritten/Rescue 36/36. Additive 35/36. Zero paired harms; one Rewritten win.	The v4 signature did not replicate. Immutable-additive mitigation not supported by this test.

312	1	0
completed direct-memory model calls	non-replicated correlated signal	repeatable material degradations meeting threshold

Experiment verdict: the tested Hunch configurations did not show repeatable material memory degradation. That result increases confidence for bounded deployment, but it does not prove the absence of rare, long-horizon, model-specific, or retrieval-mediated failures.

05 / ASSURANCE MATRIX

Confidence by safety property

Grades reflect evidence quality and scope, not just whether a test was green.

Property	Grade	Basis	Residual action
Durable write integrity	Strong	Atomic-write implementation, blocking invariant, and repository tests.	Keep fault-injection and recovery checks in CI.
Historical auditability	Strong	JSON source of truth, Git history, additive records, refusal of physical compaction.	Verify backup restore operationally.
Ambiguous truth fails safe	Strong	currentForTopic returns no authority on collision; explicit tests cover behavior.	Monitor collision and abstention rates.
Agent memory cannot silently block	Strong	Unconfirmed agent testimony is capped at advisory warning; human confirmation required.	Protect token/countersignature path.
Resistance to repeated rewriting	Moderate	v2 clean; v4 produced one correlated replacement-only signal; v5 did not replicate it.	Continue multi-model and longer-horizon tests.
Accumulated retrieval safety	Insufficient	Product pilots were floor-limited; direct tests do not fully exercise production retrieval.	Build a non-floor end-to-end retrieval benchmark.
Performance benefit	Insufficient	No completed experiment demonstrates better coding outcomes from Hunch memory.	Measure task success, latency, abstention, and correction cost.
Universal memory safety	Not established	No finite test suite or cited paper supports this universal claim.	Do not make this claim.

Repository health at evidence cutoff

Check	Result
Automated tests	1371 total: 1370 passed, 0 failed, 1 skipped
Type checking	Passed
Conformance	7/7 invariants passed
Drift	No drift reported
Evidence commit	47a7e5d (v5 report); preregistration commit 33cd648

These checks support implementation integrity and reproducibility. They do not, by themselves, establish universal behavioral safety.

Conditions that must stay true

Release is conditional on preserving the controls that make the bounded assurance credible.

Priority	Condition	Gate	Verification
P0	Authoritative JSON writes remain atomic; failure leaves the prior target intact.	Required	Build and fault-injection tests
P0	Physical compaction remains disabled until recoverable tombstone-based garbage collection is designed and validated.	Required	CLI conformance test
P0	Unconfirmed agent records remain advisory and cannot create blocking authority.	Required	Authority boundary tests
P0	Unresolved collisions fail closed: no single current answer is injected as authority.	Required	Topic collision tests and telemetry
P0	Backups and Git history are retained; restore is exercised before high-impact rollout.	Required	Operational restore drill
P1	Conformance, type checking, drift, and memory regression suites gate release.	Required	CI release gate
P1	Telemetry tracks retrieval misses, UNKNOWN/abstention, collisions, supersessions, and corrections.	Required	Production dashboard and alerts
P1	Rollback and record supersession are tested on the deployed storage path.	Required	Staging rehearsal

Deployment scope

APPROPRIATE NOW	NOT YET JUSTIFIED
- Reversible use with human review - Auditable decision and constraint memory - Staged rollout with monitoring - Fast rollback and supersession	- Irreversible or safety-critical autonomy - Unsandboxed authority from agent memory - Claims of certification or zero risk - Unmonitored large-scale memory accumulation

07 / EVIDENCE NEEDED NEXT

How to move from bounded assurance to stronger proof

The next work should target the known evidence gaps, not repeat already inconclusive protocols.

Order	Experiment	Pass evidence
1	Non-floor end-to-end retrieval benchmark	Tasks where clean baseline is neither 0% nor 100%; measure source contract, issue contract, task success, and false-authority injection.
2	Long-horizon multi-model stress	Run multiple current model families across larger heterogeneous memories and preregister rare-failure thresholds.
3	Storage fault injection	Interrupt writes, corrupt derived indexes, simulate disk-full and process crash, then verify exact recovery from authoritative JSON and Git.
4	Production canary	Roll out to a small reversible cohort; alert on collision, abstention, retrieval miss, supersession, and human-correction rates.
5	Restore and rollback drill	Demonstrate recovery-time and recovery-point objectives using the deployed storage and backup configuration.

Evidence index

ID	Evidence	Location
P1	Research paper and discussion	Public paper on arXiv Discussion on alphaXiv
E1	Consolidation-safety pilot	bench/external/results/2026-08-26-hunch-consolidation-safety-pilot.md
E2	Calibrated follow-up	bench/external/results/2026-08-27-hunch-consolidation-calibrated-followup.md
E3	Direct degradation v2	bench/external/results/2026-08-27-hunch-direct-memory-degradation-v2-report.md
E4	Capacity stress v4	bench/external/results/2026-08-27-hunch-direct-memory-capacity-stress-v4-report.md
E5	Schema collision v5	bench/external/results/2026-08-27-hunch-schema-collision-replication-v5-report.md
C1	Core implementation	src/store/jsonStore.ts; src/core/io.ts; src/core/topics.ts; src/cli/index.ts
T1	Safety tests	test/compact.test.ts; test/topics.test.ts; test/testimony.test.ts; test/veto.test.ts
D1	Retrieval design record	.hunch/decisions/dec_e622668785.json
I1	Storage invariants	.hunch/constraints/con_902759b3dc.json; con_a87360128b.json

Final assurance statement: Hunch has credible architectural safeguards and favorable bounded test evidence for controlled production use. The correct claim is 'engineered and tested for safer memory behavior,' not 'proven safe.'